

Analyzing Smart City Bike Sharing Data using Power BI

Project Objective

To analyze bike station performance, usage efficiency, and operational patterns across different locations and time periods using Power BI, and to provide data-driven insights to improve bike availability, optimize resource distribution, and enhance overall system efficiency. Create Interactive dashboards and to support decision making.

Data Preparation & Modeling Document

Overview

The dataset initially contained all information in a **single flat table**, including station details, contract information, and bike availability data. To improve analysis and performance in Power BI, the data was transformed into a **star schema model** using Power Query

Problem Statement

- Some stations have **no bikes (empty)** while others have more bikes
- **Low occupancy rate** indicates underutilization of stations
- **Uneven distribution of bikes** affects system efficiency
- **Usage patterns vary by time and month**, causing demand fluctuations
- **Performance differs across locations/contracts (cities)**
- Presence of **closed stations** impacts service availability
- Lack of proper monitoring can lead to **poor decision-making**

Initial Dataset Structure

The original dataset included the following columns:

- Number (Station Number)
- Name (Station Name)
- Address
- Position (Latitude & Longitude)
- Banking
- Bonus

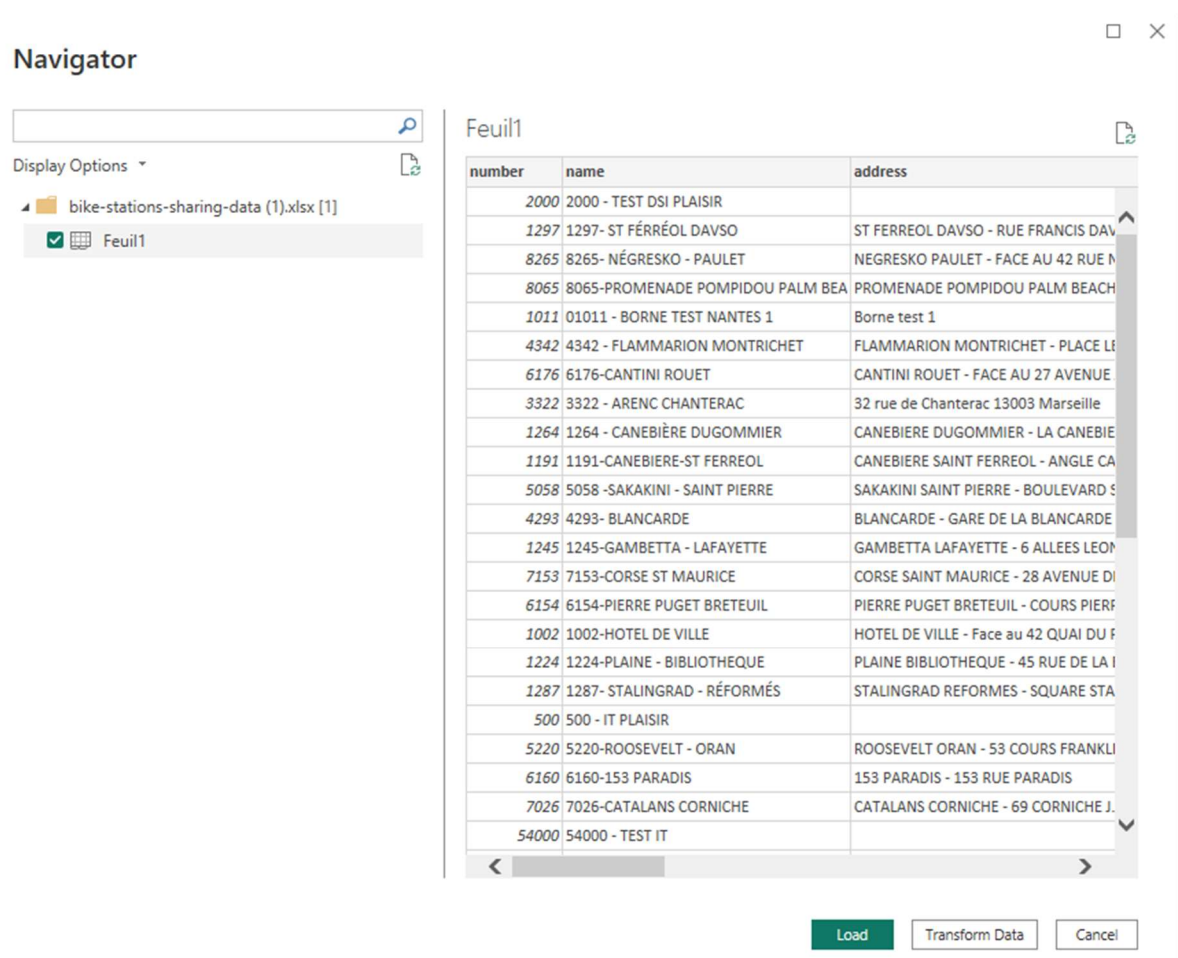
- Contract Name
- Bike Stands
- Available Bikes
- Available Bike Stands
- Last Update (Date & Time)

☞ This structure was **denormalized** and not suitable for efficient analysis.

🔄 Data Transformation (Power Query)

The following steps were performed:

- ✓ Imported dataset into **Power Query**



The screenshot shows the Power Query interface. On the left, the 'Navigator' pane displays a folder named 'bike-stations-sharing-data (1).xlsx [1]' with a sub-item 'Feuil1' selected. The main area shows the 'Feuil1' sheet as a table with three columns: 'number', 'name', and 'address'. The table contains 25 rows of data, including station numbers, names, and addresses. At the bottom right, there are three buttons: 'Load', 'Transform Data', and 'Cancel'.

number	name	address
2000	2000 - TEST DSI PLAISIR	
1297	1297- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAV
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE N
8065	8065-PROMENADE POMPIDOU PALM BEA	PROMENADE POMPIDOU PALM BEACH
1011	01011 - BORNE TEST NANTES 1	Borne test 1
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE L
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE
3322	3322 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIE
1191	1191-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CA
5058	5058-SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD S
4293	4293- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON
7153	7153-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DI
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERF
1002	1002-HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU F
1224	1224-PLAINE - BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA I
1287	1287- STALINGRAD - RÉFORMÉS	STALINGRAD REFORMES - SQUARE STA
500	500 - IT PLAISIR	
5220	5220-ROOSEVELT - ORAN	ROOSEVELT ORAN - 53 COURS FRANKLI
6160	6160-153 PARADIS	153 PARADIS - 153 RUE PARADIS
7026	7026-CATALANS CORNICHE	CATALANS CORNICHE - 69 CORNICHE J.
54000	54000 - TEST IT	

- Cleaned data by:
 - Removing duplicates
 - Handling missing values
 - Trimming text fields

Table.TransformColumnTypes({"Promoted Headers",{"number", Int64.Type}, {"name", type text}, {"address", type

	position	bonus	status	Contr
1	46.5488766782771, 15.635		FALSE	maribor
2	43.293485736186526, 5.37		FALSE	marseill
3	43.269318556371154, 5.38		FALSE	marseill
4	43.267272515671856, 5.37		FALSE	marseill
5	47.19504, -1.5575		FALSE	nantes
6	43.305993453766845, 5.39		FALSE	marseill
7	43.284775983612235, 5.38		FALSE	marseill
8	43.3123370339008, 5.3687		FALSE	marseill
9	43.29789631329341, 5.381		FALSE	marseill
10	43.29620767061392, 5.376		FALSE	marseill
11	43.292800863545835, 5.40		FALSE	marseill
12	43.29626340517055, 5.406		FALSE	marseill
13	43.29902793826475, 5.381		FALSE	marseill
14	43.28946760889984, 5.363		FALSE	marseill
15	43.290329770206334, 5.37		FALSE	marseill
16	43.2961387765145, 5.3707		FALSE	marseill
17	43.29582518327951, 5.385		TRUE	marseill
18	43.29552097558015, 5.385		FALSE	marseill
19	48.799828, 1.946855		FALSE	jcdcau
20	43.29899195650368, 5.388		TRUE	marseill
21	43.28683835122117, 5.379		FALSE	marseill
22	43.28985620147398, 5.355		FALSE	marseill
23	48.799887, 1.947001		FALSE	nancy
24				

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 17:15

- ✓ Split Date and Time from “Last Update” column

Queries [4]

fact

date_dim

contract_dim

Station_dim

Table.SelectRows("#Reordered Columns1", each true)

	Date_ID	Date	Month	year	time
1	D001	09	06		2025 13:11:37 +05:30
2	D002	31	10		2025 11:37:27 +05:30
3	D003	17	01		2024 13:59:16 +05:30
4	D004	20	05		2024 11:11:59 +05:30
5	D005	15	07		2025 13:11:26 +05:30
6	D006	04	05		2023 15:42:08 +05:30
7	D007	03	07		2023 13:27:37 +05:30
8	D008	26	03		2025 11:48:54 +05:30
9	D009	12	11		2025 14:48:33 +05:30
10	D010	12	11		2025 14:48:01 +05:30
11	D011	12	11		2025 14:51:25 +05:30
12	D012	12	11		2025 14:49:46 +05:30
13	D013	12	11		2025 14:53:53 +05:30
14	D014	12	11		2025 14:54:50 +05:30
15	D015	12	11		2025 14:51:54 +05:30
16	D016	12	11		2025 14:57:16 +05:30
17	D017	12	11		2025 14:56:50 +05:30
18	D018	12	11		2025 14:55:42 +05:30
19	D019	12	11		2025 14:57:15 +05:30
20	D020	12	11		2025 14:50:09 +05:30
21	D021	12	11		2025 14:54:49 +05:30
22	D022	12	11		2025 14:54:19 +05:30
23	D023	12	11		2025 14:55:47 +05:30
24	D024	03	07		2023 13:29:43 +05:30

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

- ✓ Extracted Latitude and Longitude from position field

Query Settings

Formula Bar: `= Table.TransformColumnTypes(Table1_Table,{{"Column1", type text}, {"position", type text}, {"banking", type`

Columns: Column1, position, banking, bonus, status

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter:

Split at:

- ☐ Left-most delimiter
- ☐ Right-most delimiter
- ☒ Each occurrence of the delimiter

Advanced options

Quote Character:

☐ Split using special characters

Insert special character:

OK Cancel

	Column1	position
1	S001	49.901646, 2.303244
2	S002	59.95363, 11.04428
3	S003	43.461719, -3.8021
4	S004	48.7988350967829, 2.4
5	S005	45.7246304232008, 4.8
6	S006	48.7999583773285, 1.9
7	S007	49.4395241490918, 1.0
8	S008	47.742424, 7.342152
9	S009	55.7065202995297, 13.
10	S010	36.705842, 137.215829
11	S011	46.051367, 14.506542
12	S012	37.4129235511391, -5.5
13	S013	39.48001, -0.38302
14	S014	49.598211, 6.137526
15	S015	50.47536, 4.84661
16	S016	43.60419, 1.445475
17	S017	47.2198398196124, -1.5
18	S018	50.836304, 4.41505
19	S019	46.562695, 15.62935
20	S020	48.6924654452072, 6.1
21	S021	53.340927, -6.262501
22	S022	47.246419, 6.022567
23	S023	54.68590826, 25.28560

Formula Bar: `= Table.RenameColumns("#Changed Type1",{{"position.1", "Latitude"}, {"position.2", "Longitude"}})`

Columns: Column1, 1.2 Latitude, 1.2 Longitude, banking, bonus, status

	Column1	1.2 Latitude	1.2 Longitude	banking	bonus	status
1	S001	49.901646	2.303244	FALSE	FALSE	OPEN
2	S002	59.95363	11.04428	FALSE	FALSE	CLOSED
3	S003	43.461719	-3.8021	TRUE	FALSE	CLOSED
4	S004	48.7988351	2.453876338	TRUE	FALSE	OPEN
5	S005	45.72463042	4.829194816	FALSE	FALSE	OPEN
6	S006	48.79995838	1.946886971	FALSE	FALSE	OPEN
7	S007	49.43952415	1.08880495	TRUE	FALSE	CLOSED
8	S008	47.742424	7.342152	TRUE	FALSE	CLOSED
9	S009	55.7065203	13.18741858	TRUE	FALSE	OPEN
10	S010	36.705842	137.215829	FALSE	FALSE	OPEN
11	S011	46.051367	14.506542	FALSE	FALSE	OPEN
12	S012	37.41292355	-5.988905938	FALSE	FALSE	OPEN
13	S013	39.48001	-0.38302	FALSE	FALSE	OPEN
14	S014	49.598211	6.137526	FALSE	FALSE	OPEN
15	S015	50.47536	4.84661	FALSE	FALSE	OPEN
16	S016	43.60419	1.445475	FALSE	FALSE	OPEN
17	S017	47.21983982	-1.554891341	TRUE	FALSE	OPEN
18	S018	50.836304	4.41505	FALSE	TRUE	OPEN
19	S019	46.562695	15.62935	FALSE	FALSE	OPEN
20	S020	48.69246545	6.183400554	TRUE	FALSE	OPEN
21	S021	53.340927	-6.262501	FALSE	FALSE	OPEN
22	S022	47.246419	6.022567	TRUE	TRUE	OPEN
23	S023	54.68590826	25.28560002	FALSE	FALSE	CLOSED

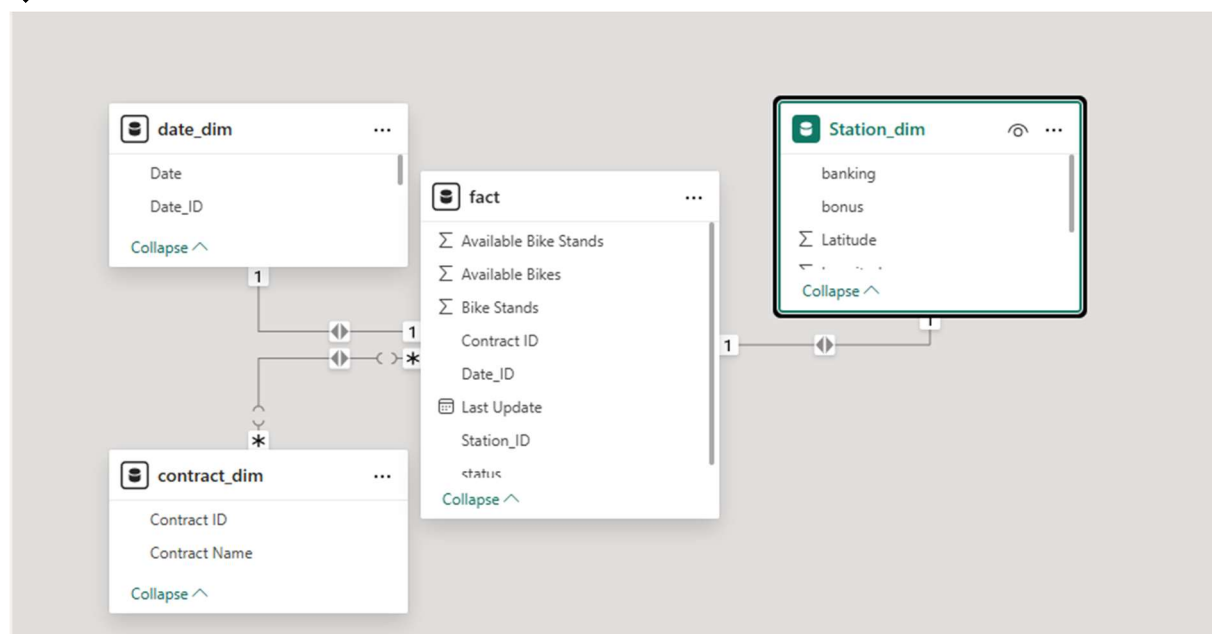
1 profiling based on top 1000 rows

- ✓ Standardized column formats

1 ² Bike Stands	1 ² Available Bike Stands	1 ² Available Bikes	Last Update	Contract Name - Copy
20	16	0	09-06-2025 13:11:37	C001
20	10	10	31-10-2025 11:37:27	C002
40	36	4	17-01-2024 13:59:16	C003
26	26	0	20-05-2024 11:11:59	C004
6	0	0	15-07-2025 13:11:26	lyon
1	0	0	04-05-2023 15:42:08	cergy-pontoise
25	2	0	03-07-2023 13:27:37	rouen
40	39	0	26-03-2025 11:48:54	mulhouse
40	23	16	12-11-2025 14:48:33	lund
16	7	9	12-11-2025 14:48:01	toyama
20	7	13	12-11-2025 14:51:25	ljubljana
20	17	3	12-11-2025 14:49:46	seville
25	21	4	12-11-2025 14:53:53	valence
20	19	1	12-11-2025 14:54:50	luxembourg
15	8	7	12-11-2025 14:51:54	namur
19	1	18	12-11-2025 14:57:16	toulouse
33	28	4	12-11-2025 14:56:50	nantes
20	8	12	12-11-2025 14:55:42	Bruxelles-Capitale
22	13	9	12-11-2025 14:57:15	maribor
25	7	18	12-11-2025 14:50:09	nancy
31	8	23	12-11-2025 14:54:49	dublin
30	16	14	12-11-2025 14:54:19	besancon
21	0	0	12-11-2025 14:55:47	vilnius

Data Modeling Approach

The dataset was transformed into a **Star Schema** consisting of:



1. Fact Table

Contains transactional and measurable data:

- Station ID
- Contract ID
- Date ID
- Available Bikes
- Available Bike Stands
- Bike Stands
- Status

2. Station Dimension Table

Contains station-related descriptive data:

- Station ID (*manually created*)
- Station Name
- Address
- Latitude
- Longitude
- Banking
- Bonus

3. Contract Dimension Table

Contains contract/city information:

- Contract ID (*manually created*)
- Contract Name

4. Date Dimension Table

Created from date-time field:

- Date ID (*manually created*)
- Date
- Month
- Year
- Time

ID Creation (Important Step)

Since the dataset did not contain unique identifiers:

- Station ID, Contract ID, and Date ID were **manually generated**

- These IDs were used to:
 - Establish relationships
 - Maintain data integrity
 - Enable efficient filtering and aggregation

Relationships (Star Schema)

- Fact Table → Station Dimension (Station ID)
- Fact Table → Contract Dimension (Contract ID)
- Fact Table → Date Dimension (Date ID)

Dax and Measures used in this project:

KPI Cards Measures:

Total Bike Available:

Total Bikes = SUM(Fact_Table[Available Bikes])

Total Bike Stands:

Total Stands = SUM(FactTable[Bike Stands])

Available Bike Stands:

Available Stands = SUM(FactTable[Available Bike Stands])

Occupancy Rate:

```
Occupancy Rate =
DIVIDE(
    SUM(FactTable[Available Bikes]),
    SUM(FactTable[Bike Stands]),
    0
)
```

Empty Stations

```
Empty Stations =
CALCULATE(
    COUNT(FactTable[Station ID]),
    FactTable[Available Bikes] = 0
)

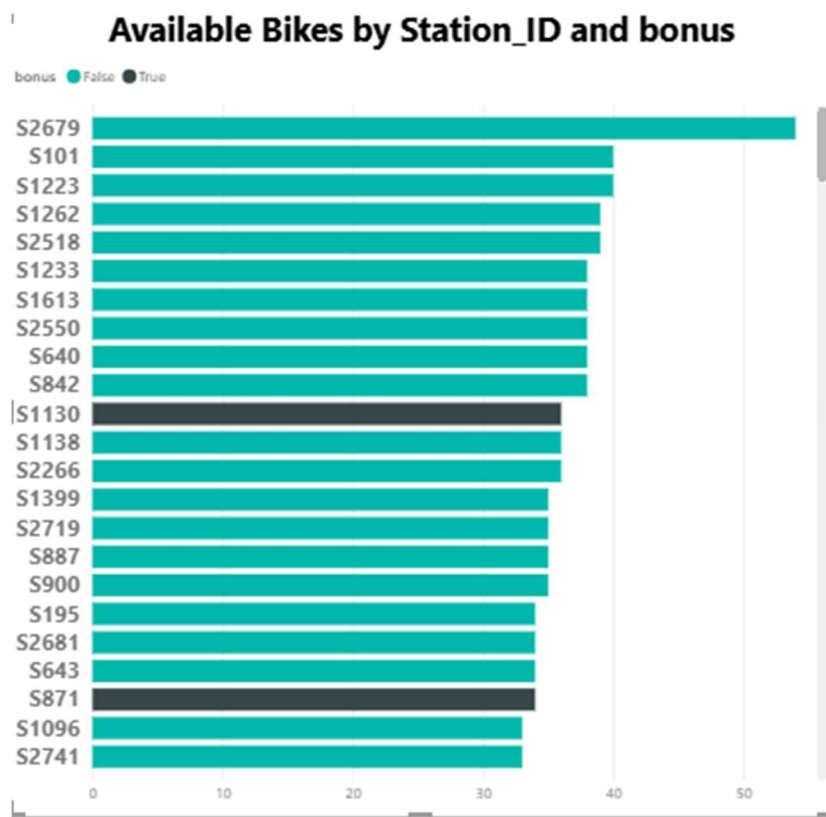
Full Stations
Station Rank =
RANKX(
    ALL(Station_dim[Station ID]),
    [Total Bikes],
    ,
    DESC
)
```

Top Stations (Rank)

```
Full Stations =  
CALCULATE(  
    COUNT(FactTable[Station ID]),  
    FactTable[Available Bike Stands] = 0  
)
```

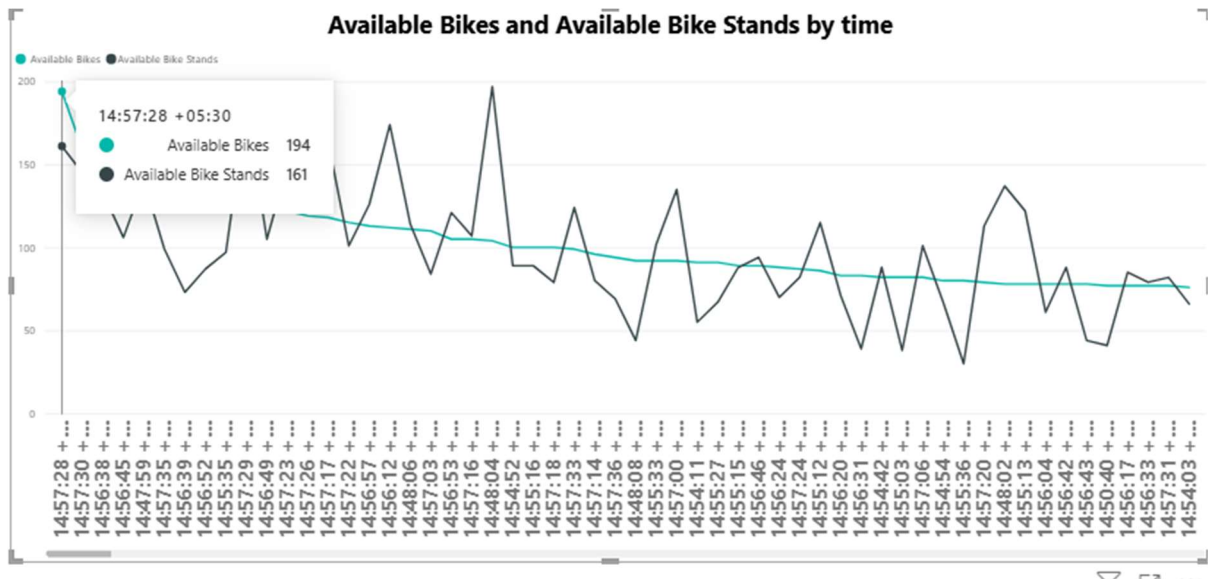
Important Visualization:

- Highest Bike stands available in this city,

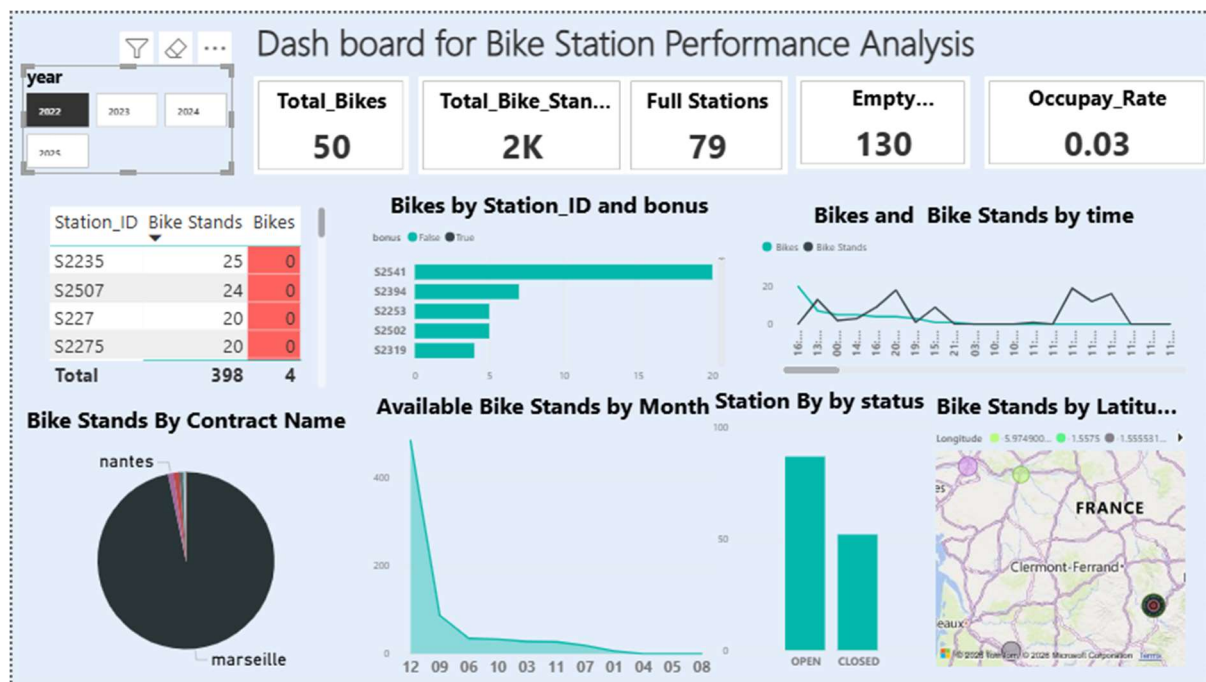


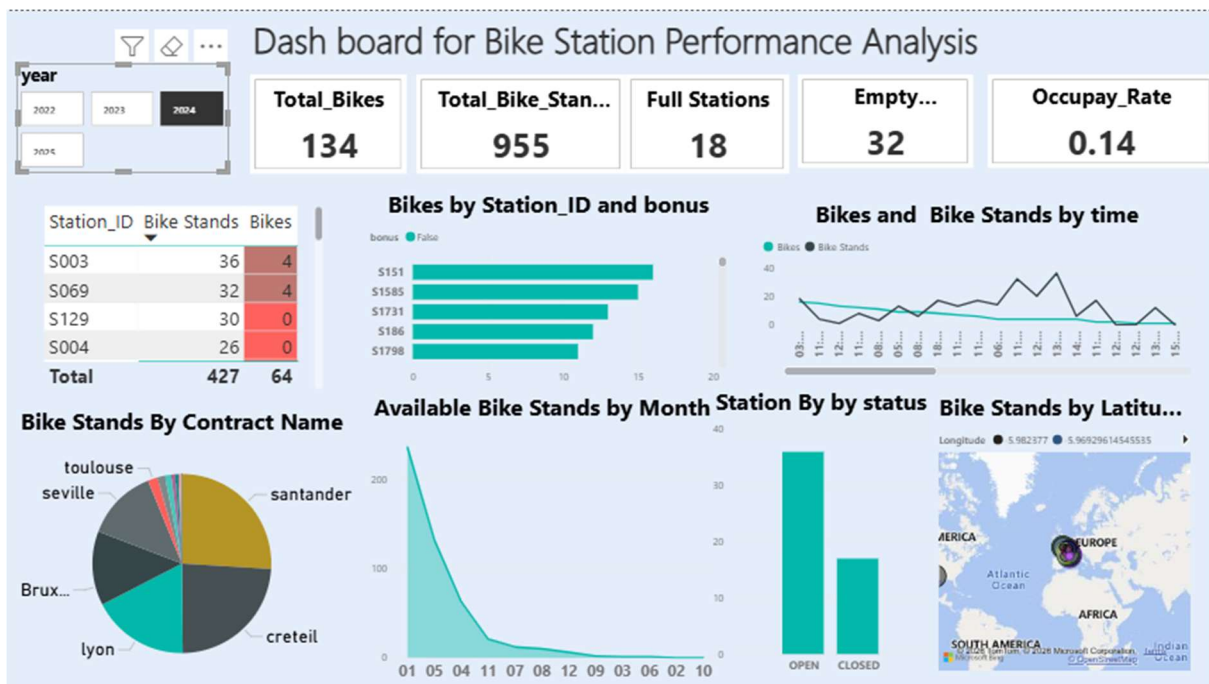
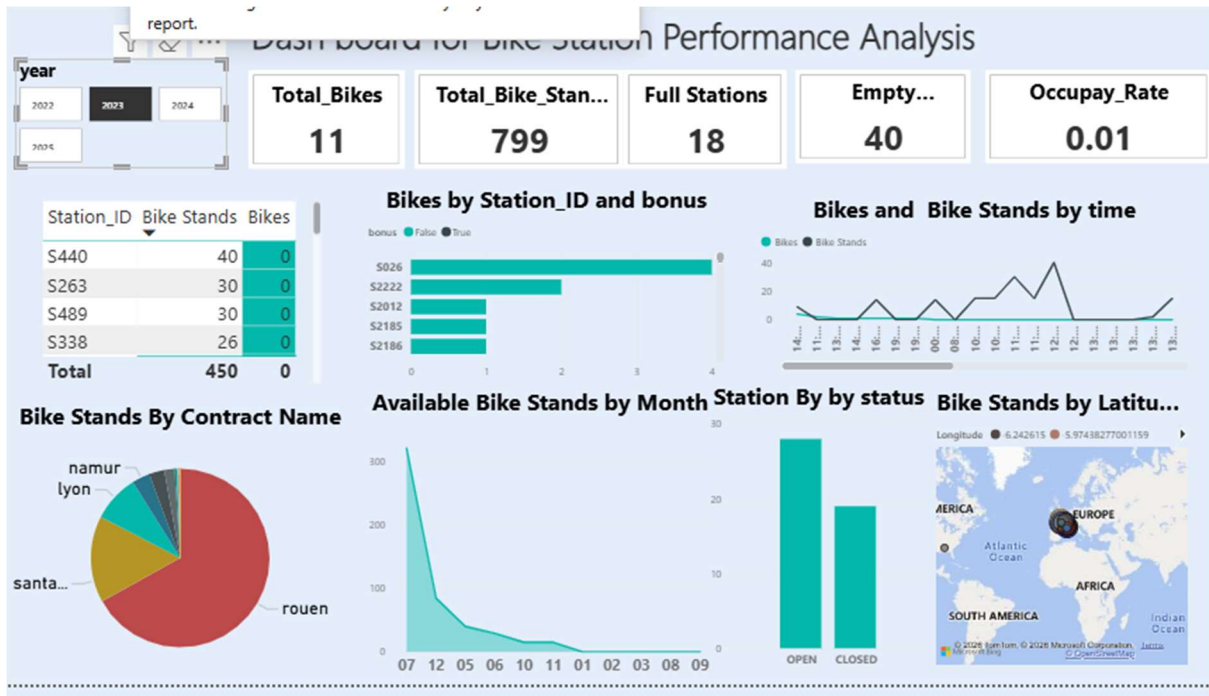
Key Insights:

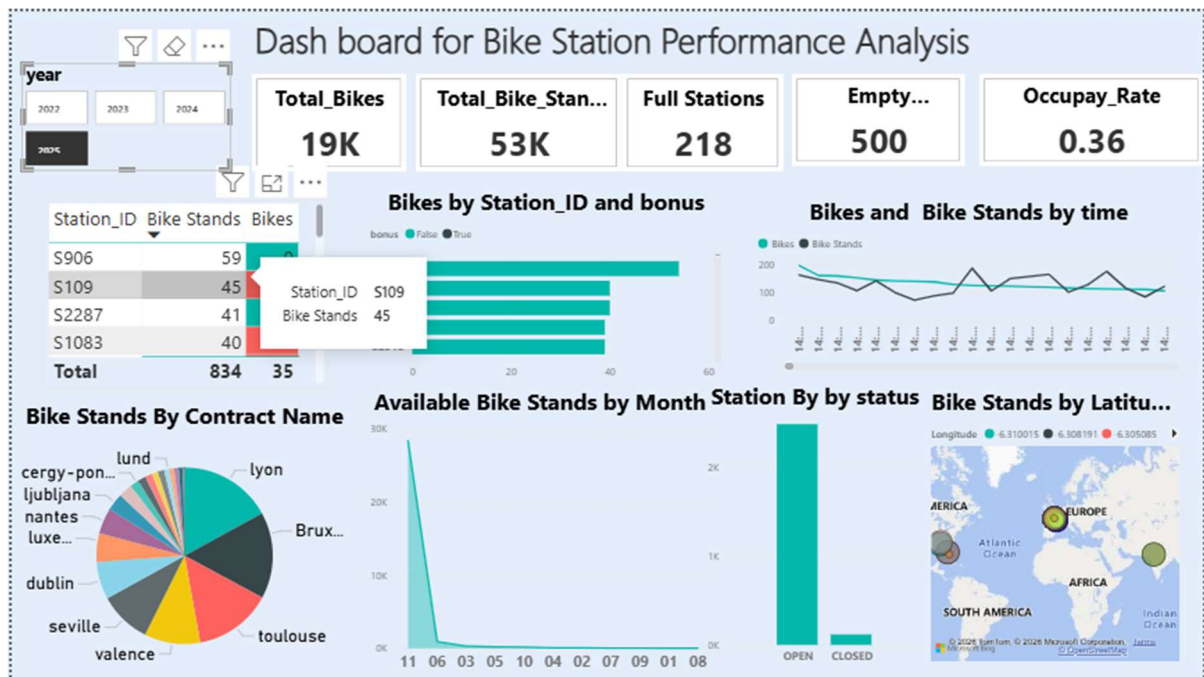
- Few stations have bonus Points



FINAL DASHBOARD :







4 Types of Analysis with Real-Time Examples

1. Descriptive Analysis :

- Total Bikes = ~19K
- Empty Stations = ~500
- Occupancy Rate = 0.36

“The dashboard shows current system status, including total bikes, empty stations, and occupancy rate.”

2. Diagnostic Analysis

- Some stations show **0 bikes (highlighted in red)**
- Low occupancy stations identified
- More Demand in November Month

“Stations with zero bikes indicate high demand or poor bike distribution, which explains the imbalance in the system.”

3. Predictive Analysis

- **Time chart (Bikes vs Time)**
- **Monthly trend chart**

- **“Based on time trends, bike usage increases during peak hours, so similar demand is expected in the future.”**

◆ 4. Prescriptive Analysis

- Low occupancy stations
- Bonus & banking data

“To improve the system, bikes can be redistributed to empty stations and more bonus and banking stations can be added to increase usage.”

Conclusion:

“This project successfully analyzes bike sharing data using Power BI to understand station performance, usage efficiency, and operational patterns. The dashboard highlights key issues such as uneven bike distribution, low occupancy at certain stations, and varying demand across time and locations. By using data visualization and DAX measures, meaningful insights were generated. Based on the analysis, strategies such as redistributing bikes and increasing bonus and banking stations are recommended to improve system efficiency and user satisfaction.”